Journal of Science and Medicine in Sport xxx (xxxx) xxx

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Original investigation

Tackle and ball carrier demands of rugby league: a seven-year league-wide study including over 1,000,000 tackle events

Gordon Rennie [a,b,*], Dan Weaving [a,c], Brian Hart [b], Nicholas Dalton-Barron [a,e], Ben Jones [a,c,d,f,g]

[a] Carnegie Applied Rugby Research (CARR) Centre, Carnegie School of Sport, Leeds Beckett University, United Kingdom
[b] Catapult Sports, Australia
[c] Leeds Rhinos Rugby League Club, United Kingdom
[d] England Performance Unit, Rugby Football League, Red Hall, United Kingdom
[e] Department of Sports Science and Research Innovation, PlayerMaker™, UK
[f] School of Science and Technology, University of New England, Australia
[g] Division of Exercise Science and Sports Medicine, Department of Human Biology, Faculty of Health Sciences, the University of Cape Town and the Sports Science Institute of South Africa, South Africa

ARTICLE INFO

Article history:
Received 2 June 2022
Received in revised form 5 July 2022
Accepted 6 July 2022
Available online xxxx

Keywords:
Rugby league
Tackle
Team sports
Match demands

ABSTRACT

Objectives: Describe the highest frequency and variability for tackle events in rugby league. Investigate seasonal differences in total tackle events per match over a seven-year period.
Design: Retrospective observational.
Methods: Tackle events (i.e., ball carrier events [attacker] and tackler involvements [defender]) from 864 male professional rugby league players competing in 1176 Super League matches from 2014 to 2020 were included. A series of linear mixed effect models were used to determine the frequency and variability during peak 1-, 3-, 5-, 10-, 20-, 40-min and whole-match tackle events per player per match at a positional group level. Differences between seasons for the total number of tackle events per match were compared using a one-way analysis of variance and with Tukey's honestly significant difference test.
Results: Tackle events were greatest for Props (51.5 [47.7–55.4] per match). Within-players, between-matches, and between-seasons variability was <10 % for tackle events. There were significantly less tackle events and tackler involvements per match in 2014 and a significantly more tackle events per match in season 2020_b when compared with all other seasons.
Conclusions: Large between-position variability in peak tackle events, ball carrier events, and tackler involvements would suggest that coaches should separate players into positional groups and prescribe training accordingly. Total number of tackle events, ball carrier events, and tackler involvements were significantly greater in season 2020_b when compared to season 2014 to 2019 (inclusive) which may be a consequence of rule changes introduced to the sport.
© 2022 Sports Medicine Australia. Published by Elsevier Ltd. All rights reserved.

Practical implications

- Understanding the exposure and frequency of tackle events, ball carries, and tackler involvements is paramount for practitioners to adequately prepare players to the demands of the rugby league match-play.
- Large between-position variability in peak tackles, carries, and tackler involvements indicates players in different positional groups should have bespoke training, playing and recovery schedules.
- Given the stable between-match variability within a positional group, practitioners should monitor for meaningful increases and decreases in tackle event exposure between matches and plan subsequent training accordingly.
- Total number of tackles, carries, and tackler involvements were significantly greater in season 2020_b when compared to season 2014 to 2019 which may be a consequence of rule changes introduced to the sport ahead of competitive return following COVID-19 restrictions.

1. Introduction

Rugby league players undertake intermittent locomotor activities (e.g., accelerating and sprinting) and tackle events (i.e., ball carrier [attacker] and tackler involvement [defender]) during match play.[1–3] The tackle event is the most injurious activity.[4] The characteristics (e.g., frequency, intensity, and duration) of tackle events are typically

* Corresponding author.
E-mail address: g.rennie@leedsbeckett.ac.uk (G. Rennie).
@RennieG87

https://doi.org/10.1016/j.jsams.2022.07.002
1440-2440/© 2022 Sports Medicine Australia. Published by Elsevier Ltd. All rights reserved.

reported using microtechnology or video-based notational analyses.[5] Locomotor activities are well documented[1,6,7] yet the characteristics pertaining to tackle events are less understood, despite their importance from a physical performance and an injury perspective.[5]

Tackle events are typically reported as total frequency (i.e., tackle events per match) or rate (i.e., tackle events per minute).[3,8] Understanding the frequency and rate of tackle events collectively, and separately as a ball carrier or involvement as a tackler, is important from a load monitoring, injury risk reduction, and physical preparation perspective.[1,5,9] During rugby league matches, forwards are involved in a similar number of tackle events as a ball carrier ($n = 11.2$ vs 11.4 per match) and approximately twice as many involvements as a tackler ($n = 24.6$ vs 12.8 per match) compared to backs.[5] Whilst useful, these data are based on whole-match frequencies, and do not report the maximal number of tackle events a player is involved in for a given duration (e.g., 1-min, 5-min, or 10-min) which may impact both performance and injury risk within a match.

Johnston et al.[2] used a rolling sum approach to determine the peak tackle event rate (i.e., tackle events per minute) for forwards and backs in Super League (SL) and National Rugby League (NRL), which are the two professional rugby league competitions based in Europe and Australasia, respectively. The peak number of tackle events per minute for a 1-min epoch were the same for SL and NRL forwards and backs (e.g., 4 ± 1 and 3 ± 1 tackle events per minute). Weaving et al.[3] reported that SL players were exposed to 4.9 to 9.7 tackle events during their highest average running demands during a 10-min period. Both of these studies[2,3] measured tackle events using a validated microtechnology collision detection algorithms,[10] which does not differentiate if a players involvement was a ball carrier (i.e., attacker) or tackler (i.e., defender). Consequently, the peak ball carry and tackle demands of rugby league match-play remain unknown and requires investigation. This is important given 61 % of injuries during rugby league occur during the tackle event,[11] and the ball carrier has a higher injury rate than the tackler.[4]

Rugby league has also implemented several rule changes. In 2020, SL temporarily removed the scrum and introduced the 'six-again' rule. In comparison to the previous season, adjustables and backs completed more ball carries, and forwards completed more tackles during matches.[12] In the NRL, following rule changes the mean duration of a ball-in-play phase increased from 76.6 ± 11.9 s in 2019 to 90.1 ± 14.9 s and 98.7 ± 17.5 s in 2020 and 2021,[13] suggesting more tackles are undertaken prior to a rest period (i.e., ball out of play). Whilst research has shown a difference in overall match demands due to rule changes,[12,13] the evolution of the total number of tackle events, and tackle events as a ball carrier or tackler or per match, and for different durations of time (e.g., 1- to 10-min) is yet to be determined.

The primary aim of this study was to describe the highest frequency and variability of tackle events, ball carries and tackler involvements per player for playing positions during professional rugby league matches over a seven-year period. The secondary aim was to investigate seasonal differences in total (i.e., sum for all players during a match) tackle events, and involvements as a ball carrier or tackler during rugby league matches between 2014 and 2020.

2. Materials and methods

This study used a retrospective observational design in which tackle events, ball carries, and tackler involvements were analysed from 864 individual male professional rugby league players competing in 1176 SL fixtures from 2014 to 2020. Eight observational periods were included: one for each season (2014–2019) and two for the 2020 season (2020_a pre-COVID break and 2020_b post-COVID break). Season 2020 was separated into two sub-seasons due to the duration of the enforced break in fixtures[12] and different rules of match-play (e.g., with and without scrums[14]). Players were categorised by their primary playing position during each match; props ($n = 315$), hooker ($n = 169$), second rows ($n = 262$), loose forward ($n = 292$), scrum half ($n = 130$), stand-off ($n = 149$), centres ($n = 229$), wings ($n = 216$) and fullback ($n = 129$). A breakdown of the number of fixtures and positional groups during each match are available in Supplementary Materials, Table 1. Institutional ethics approval was granted by Leeds Beckett University Human Ethics Research Committee.

Match-play event data was provided by Opta (Stats Perform, London, United Kingdom) and extracted online (https://www.optaprorugby.com/index.php) as an extensible markup language (XML) file. Commercial match-event providers have been used extensively as a data source within rugby league research.[11,15–18] Each XML file was converted to a comma separated value (CSV) file that contained date, a unique fixture identification number (ID), player ID, position category, event ID, event timestamp, and event type (e.g., ball carry or tackler involvements). *Ball carries* were defined as a player being in possession of the ball when being tackled by a defending player and included instances whereby the ball carrier offloaded the ball in the process of being tackled.[16,19] *Tackler involvements* were defined as a player attempted to halt the progress or dispossess an opponent in possession of the ball regardless of the outcome (e.g., incomplete tackles during which the defender made initial contact but 'missed' the tackle due to physical contact with the attacker were included within the tackler involvement frequency[19]). *Tackle events* were interpreted as collective ball carries and tackler involvements. These operating definitions[19] were consistent throughout the data collection period.

A CSV file containing the precise second of tackle events, as a ball carrier or involvement as a tackler, for each player and match was imported into R (version 4.1.2) for analysis. The highest number of tackle events, ball carries and tackler involvements accumulated per individual within each match were calculated for each defined time epoch (e.g., 1, 3, 5, 10, 20, 40-min and whole-match) using a rolling sum via the *roll_sum()* function, contained within the *RccpRoll* package.[20] The highest value was taken as the peak tackle event frequency for each time epoch.[2,3] The total number of tackle events, ball carries, and tackler involvements performed within each match (i.e., sum total of events during a match) were determined by grouping and summating all defined events within a match.

Prior to analysis, normality was inspected through kernel density and quantile-quantile plots. The distributions of peak match-event dependant variables (tackle events, ball carries, and tackler involvements) were not normal, so variables were log-transformed to reduce error arising from non-uniform residuals and then back-transformed post-analysis. To account for zero values within the peak match-event dataset, a value of 1 was added to each count variable which was subsequently deducted from the resulted back transformed estimates. The distribution of total tackle events, ball carries, and tackler involvements per match was normally distributed.

For the peak 1-, 3-, 5-, 10-, 20-, 40-min and whole-match analysis, peak match-event data were analysed through a series linear mixed effects models (LMM's) using the *lme4* package in R,[21] with random effects for player ID (to estimate differences between-player means), position category (to estimate changes between-positions), fixture ID (to estimate changes between-matches) and season (to estimate changes between-seasons).[22] The resultant estimates were presented with 95 % confidence intervals (CI). Within-player variation was then attributed to that of otherwise unexplained variability (i.e., the residual). Variability was expressed using the coefficient of variation (CV) and was presented with 95 % CIs as markers of uncertainly estimates; large changes were interpreted as >10 %.[23]

For the total number of tackle events per match analysis, differences between the total number of events per match were compared using a one-way analysis of variance (ANOVA) using the *aov()* function in base R. Tukey's honestly significant difference (HSD) test was used post-hoc using the *TukeyHSD()* function in base R to compare differences between seasons and adjusted *p*-values were interpreted at an alpha-level of $p < 0.05$. The adjusted p-value identifies positional

comparisons that are significantly different, whilst limiting the family error rate.[24]

3. Results

A total of 1,173,014 tackle events (355,724 ball carries and 817,290 tackler involvements) were coded from 2014 to 2020 and included within this study. Table 1 presents the mean peak tackle events, ball carries, and tackler involvements per player for playing position for each time epoch.

Table 2 details the variability in tackle events, ball carries, and tackler involvements within- and between-players, between-matches, between-positions, and between-seasons. Within-players, variability (%CV) ranged from 0.5 % to 5.7 % for all time epochs. Between-matches variability ranged from 1.9 % to 8.4 % for tackle events, 1.5 % to 7.7 % for ball carries, and 2.0 % to 7.2 % for tackler involvements. Between-seasons, variability ranged from 0.6 % to 1.8 % for tackle events, 0.5 % to 2.2 % for ball carries, and 0.8 % to 1.2 % for the tackler. Between-players, variability was <10 % for tackle events (1- to 10-min epoch), ball carries (1- to 3-min epoch) and for tackler involvements (1- to 5-min epoch). Between-positions, variability was <10 % for ball carry exposures in the 1-min time epoch. All other measures of variability were >10 % CV (Table 2).

There was a significantly lower number of tackle events and tackler involvements per match in 2014 when compared with all other seasons. Except for ball carries in the 2020_b to 2020_a comparison, there was a significantly greater number of tackle events, ball carries, and tackler involvements per match in season 2020_b than all other seasons (Table 3).

4. Discussion

The primary aim of this study was to describe the highest frequency of tackle events experienced for a given duration for ball carries, and tackler involvements per player during professional rugby league matches over a seven-year period. During a whole-match, Props were involved in the highest number of tackle events (18.5 ± 95 % CI; 16.9–20.1) and ball carries (14.5; 12.6–16.4), and Hookers were involved in the highest number of involvements as a tackler (38.7; 36.7–40.7). Within a 1-min period both Props and Loose Forwards were involved in the highest number of tackle events (both 3.4; 3.3–3.5), Props and Fullbacks were involved in the highest number of ball carries (both 1.6; 1.5–1.7) per minute, and Props, Hookers and Loose Forwards were involved in the highest number of involvements as a tackler (3.2; 3.1–3.3, 3.2; 3.1–3.4, and 3.2; 3.1–3.4, respectively) per minute. Variability in peak tackle, ball carries, and tackler involvements across all time epochs was relatively stable (<10 % CV) between matches and seasons. A secondary aim was to investigate seasonal differences at a match level between 2014 and 2020, where significantly more tackles and tackler involvements per match were observed for season 2020_b, in comparison to all other seasons, which may be a consequence of competition rule changes. These findings are important considerations from a player welfare, physical preparation, and competition structure perspective.

Based on over 1,000,000 tackle events over seven years, this study showed that during a whole-match, forwards undertake 32 to 39 tackles and 6 to 15 ball carries, whilst backs undertake 7 to 21 tackles and 6 to 12 ball carries. This equates to approximately one tackle every 2 to 3 mins and one ball carry every 6 to 13 mins for forwards, or approximately one tackle every 4 to 11 mins and one ball carry every 7 to 13 mins for backs. Evans et al.[16] reported similar frequencies per positional group in SL, with forwards undertaking one tackle every approximately 2 to 3 mins and one ball carry every 5 to 7 mins, and backs undertaking approximately one tackle and one ball carry every 9 mins. During peak 1-min periods, forwards were involved in one tackle every 19 s, or one ball carry every 38 to 55 s, whilst backs was one tackle every 22 to 40 s, or one ball carry every 38 to 50 s. These data can be used to ensure players are appropriately prepared, mitigating the likelihood of reduced tackling technique due to fatigue[25] potentially increasing the risk of injury.[26] This is important since the ball carrier is almost twice as likely to be injured than the tackler (39 vs 20 injuries per 1000 h) during a tackle.[4]

This study showed the peak number of tackle events, ball carries, and tackler involvement across all time epochs between matches and seasons were relatively stable (<10 % CV). Larger variability was observed for 1-min peak ball carries (CV = 7.4 % (4.6–12.3 %)), between-position variability was large (>10 % CV) for all positional

Table 1
Peak tackle events, ball carries and tackler involvements per player for positions (Mean ± 95 % CIs).

Event	Positional group	1-min	3-min	5-min	10-min	20-min	40-min	Whole match
Tackle events (n)	Props	3.4 (3.3–3.5)	5.4 (5.3–5.6)	7.1 (6.8–7.3)	10.4 (10.1–10.8)	15.6 (15.0–16.2)	22.9 (21.9–23.8)	51.5 (47.7–55.4)
	Hookers	3.3 (3.2–3.5)	5.1 (4.9–5.3)	6.5 (6.2–6.8)	9.4 (9.0–9.8)	14.1 (13.4–14.7)	21.3 (20.3–22.3)	44.8 (42.5–47.0)
	Second rows	3.3 (3.1–3.4)	5.0 (4.8–5.1)	6.4 (6.1–6.6)	9.3 (8.9–9.6)	13.9 (13.3–14.5)	21.8 (20.9–22.6)	42.0 (40.2–43.7)
	Loose forwards	3.4 (3.3–3.5)	5.4 (5.2–5.6)	7.0 (6.7–7.2)	10.3 (9.9–10.7)	15.6 (15.0–16.2)	23.2 (22.3–24.1)	49.1 (47.0–51.2)
	Scrum-halves	2.7 (2.6–2.8)	3.9 (3.7–4.0)	4.8 (4.5–5.0)	6.8 (6.4–7.2)	10 (9.4–10.6)	14.9 (14.0–15.9)	27.4 (25.7–29.2)
	Stand-offs	2.7 (2.6–2.8)	3.8 (3.7–4.0)	4.8 (4.5–5.0)	6.8 (6.4–7.2)	10 (9.4–10.6)	15.0 (14.0–15.9)	27.6 (25.9–29.3)
	Centres	2.7 (2.6–2.8)	3.9 (3.8–4.0)	4.9 (4.7–5.0)	6.9 (6.7–7.2)	10.2 (9.8–10.6)	15.3 (14.7–15.9)	28.4 (27.3–29.6)
	Wings	2.1 (2.0–2.2)	2.9 (2.8–3.1)	3.6 (3.3–3.8)	5.0 (4.6–5.3)	7.1 (6.6–7.7)	10.3 (9.4–11.1)	18.5 (16.9–20.1)
	Fullbacks	2.1 (2.0–2.3)	3.0 (2.8–3.2)	3.7 (3.5–3.9)	5.2 (4.8–5.6)	7.6 (7.0–8.2)	11.1 (10.1–12.0)	20.3 (18.6–22.0)
Ball carries (n) (i.e., attacker)	Props	1.6 (1.5–1.7)	2.3 (2.1–2.4)	2.7 (2.6–2.9)	3.6 (3.5–3.8)	4.9 (4.6–5.1)	6.7 (6.3–7.1)	14.5 (12.6–16.4)
	Hookers	1.1 (1.1–1.2)	1.5 (1.3–1.6)	1.7 (1.5–1.8)	2.1 (1.9–2.3)	2.8 (2.5–3.1)	3.7 (3.2–4.1)	6.1 (5.1–7.2)
	Second rows	1.5 (1.4–1.6)	2.0 (1.9–2.1)	2.4 (2.3–2.6)	3.2 (3.0–3.4)	4.3 (4.0–4.5)	6.0 (5.6–6.4)	10.5 (9.7–11.3)
	Loose forwards	1.5 (1.4–1.6)	2.1 (2.0–2.2)	2.5 (2.4–2.6)	3.3 (3.1–3.5)	4.4 (4.2–4.7)	6.0 (5.6–6.4)	10.7 (9.7–11.7)
	Scrum-halves	1.2 (1.1–1.3)	1.6 (1.4–1.7)	1.8 (1.6–1.9)	2.3 (2.1–2.5)	3 (2.7–3.3)	4.0 (3.6–4.4)	6.4 (5.5–7.2)
	Stand-offs	1.3 (1.2–1.4)	1.7 (1.6–1.8)	1.9 (1.8–2.1)	2.4 (2.2–2.6)	3.2 (3.0–3.5)	4.3 (3.9–4.7)	6.9 (6.0–7.7)
	Centres	1.5 (1.4–1.5)	2.0 (1.9–2.1)	2.4 (2.3–2.5)	3.1 (3.0–3.2)	4.2 (4.0–4.3)	5.8 (5.6–6.1)	10.0 (9.5–10.5)
	Wings	1.5 (1.4–1.5)	2.1 (2.0–2.2)	2.5 (2.4–2.6)	3.3 (3.1–3.5)	4.6 (4.3–4.8)	6.5 (6.1–6.9)	11.3 (10.6–12.0)
	Fullbacks	1.6 (1.5–1.7)	2.2 (2.1–2.4)	2.7 (2.5–2.8)	3.5 (3.3–3.7)	4.9 (4.6–5.2)	7.0 (6.6–7.4)	12.2 (11.4–13.0)
Tackler involvements (n) (i.e., defender)	Props	3.2 (3.1–3.3)	4.8 (4.6–4.9)	6.0 (5.8–6.2)	8.5 (8.2–8.9)	12.1 (11.5–12.6)	17.3 (16.5–18.2)	37.3 (33.8–40.8)
	Hookers	3.2 (3.1–3.4)	4.8 (4.6–4.9)	6.0 (5.7–6.2)	8.5 (8.1–8.9)	12.5 (11.9–13.1)	18.7 (17.8–19.6)	38.7 (36.7–40.7)
	Second rows	3.1 (3.0–3.2)	4.4 (4.2–4.6)	5.5 (5.3–5.7)	7.8 (7.4–8.1)	11.2 (10.7–11.7)	16.9 (16.1–17.7)	31.5 (30.1–33.0)
	Loose forwards	3.2 (3.1–3.4)	4.8 (4.6–5.0)	6.1 (5.8–6.3)	8.7 (8.3–9.0)	12.6 (12.1–13.1)	18.4 (17.6–19.2)	38.4 (36.6–40.3)
	Scrum-halves	2.6 (2.4–2.7)	3.5 (3.3–3.7)	4.2 (4.0–4.5)	5.8 (5.4–6.2)	8.3 (7.7–8.8)	11.9 (11.1–12.7)	20.8 (19.3–22.4)
	Stand-offs	2.5 (2.4–2.7)	3.4 (3.3–3.6)	4.2 (4.0–4.4)	5.7 (5.4–6.1)	8.1 (7.5–8.6)	11.7 (10.9–12.5)	20.6 (19.1–22.0)
	Centres	2.4 (2.4–2.5)	3.2 (3.1–3.3)	3.9 (3.8–4.0)	5.3 (5.1–5.5)	7.4 (7.0–7.7)	10.5 (10.0–11.0)	18.4 (17.5–19.3)
	Wings	1.5 (1.4–1.6)	1.8 (1.7–2.0)	2.1 (1.9–2.3)	2.7 (2.3–3.0)	3.5 (2.9–4.0)	4.5 (3.7–5.2)	7.1 (5.8–8.5)
	Fullbacks	1.7 (1.6–1.8)	2.1 (1.9–2.3)	2.4 (2.2–2.7)	3.1 (2.7–3.5)	4.0 (3.5–4.6)	5.1 (4.3–5.9)	8.0 (6.6–9.4)

Table 2
The variability (CV ± 95 % CIs) in tackle events, ball carries and tackler involvements per player for positions.

Event	Time epoch	Residual (within-player)	Between-match	Between-season	Between-player	Between-position
Tackle events	1-min	0.66 (0.65–0.66)	1.85 (1.57–2.13)	0.57 (0.18–1.29)	5.21 (4.83–5.63)	15.38 (9.45–26.23)
	3-min	0.94 (0.93–0.94)	3.11 (2.86–3.38)	0.99 (0.51–2.03)	7.06 (6.58–7.59)	21.70 (13.19–37.64)
	5-min	1.18 (1.17–1.18)	3.88 (3.62–4.16)	1.31 (0.72–2.64)	8.11 (7.56–8.70)	25.14 (15.21–44.05)
	10-min	1.66 (1.65–1.68)	4.91 (4.62–5.22)	1.90 (1.09–3.78)	10.16 (9.50–10.89)	29.40 (17.68–52.14)
	20-min	2.54 (2.52–2.56)	5.94 (5.61–6.29)	2.83 (1.68–5.57)	12.61 (11.81–13.49)	32.85 (19.65–58.83)
	40-min	3.62 (3.60–3.65)	6.73 (6.35–7.13)	3.03 (1.79–5.98)	13.44 (12.58–14.39)	35.73 (21.28–64.47)
	Whole match	5.65 (5.59–5.71)	8.44 (7.93–8.98)	1.84 (0.85–3.93)	16.43 (15.16–17.83)	46.61 (27.24–86.52)
Ball carries (i.e., attacker)	1-min	0.54 (0.53–0.54)	1.46 (0.91–1.89)	0.48 (0.14–1.09)	6.89 (6.40–7.44)	7.36 (4.55–12.30)
	3-min	0.64 (0.64–0.65)	1.89 (1.44–2.28)	1.04 (0.60–2.05)	9.20 (8.56–9.91)	11.60 (7.15–19.61)
	5-min	0.77 (0.76–0.77)	2.32 (1.89–2.72)	1.10 (0.63–2.18)	11.16 (10.40–12.00)	13.78 (8.46–23.43)
	10-min	1.01 (1.01–1.02)	3.09 (2.66–3.51)	1.55 (0.92–3.05)	14.39 (13.44–15.44)	17.15 (10.49–29.46)
	20-min	1.37 (1.36–1.38)	4.11 (3.66–4.56)	1.83 (1.07–3.63)	18.20 (17.01–19.51)	20.57 (12.51–35.68)
	40-min	1.80 (1.79–1.81)	5.22 (4.71–5.74)	2.13 (1.23–4.23)	20.45 (19.10–21.94)	26.19 (15.80–46.13)
	Whole Match	2.92 (2.89–2.95)	7.67 (6.91–8.43)	2.22 (1.05–4.70)	26.65 (24.58–28.97)	38.15 (22.37–69.63)
Tackler involvements (i.e., defender)	1-min	0.69 (0.68–0.69)	1.99 (1.63–2.32)	0.78 (0.38–1.63)	6.44 (5.96–6.95)	22.85 (13.87–39.80)
	3-min	0.99 (0.98–1.00)	2.71 (2.36–3.07)	0.98 (0.49–2.03)	8.72 (8.11–9.38)	32.84 (19.64–58.76)
	5-min	1.24 (1.23–1.25)	3.35 (2.99–3.71)	1.16 (0.59–2.41)	10.08 (9.39–10.84)	39.03 (23.13–70.96)
	10-min	1.76 (1.74–1.77)	4.14 (3.75–4.53)	1.56 (0.85–3.17)	12.56 (11.71–13.50)	48.63 (28.43–90.56)
	20-min	2.62 (2.59–2.63)	4.76 (4.33–5.20)	2.14 (1.22–4.29)	15.27 (14.26–16.39)	59.05 (34.07–112.90)
	40-min	3.57 (3.55–3.60)	5.47 (4.95–6.00)	2.00 (1.10–4.07)	16.73 (15.59–17.98)	73.00 (41.38–144.10)
	Whole match	5.46 (5.40–5.52)	7.20 (6.26–8.13)	1.24 (0.29–2.86)	20.36 (18.62–22.32)	104.39 (57.01–220.39)

groups at all time intervals. The amount of variability in tackler events, tackles and carries appear similar to those reported for displacement variables (e.g., distance [m], average speed [$m \cdot min^{-1}$] and average acceleration [$m \cdot s^{-2}$]) in rugby league.[23] Collectively, these finding can be used to determine if a worthwhile change in tackle events or involvements has taken place, which is important for player management strategies. These findings are consistent with the unique attacking and defensive responsibilities for playing positions during match-play. Therefore, coaches and governing bodies should ensure this is reflected in player rest and recovery strategies and policies.

This study is the first to present summated tackle events data per match. There were no significant differences between any pairwise comparison from 2015 to 2020_a (pre-COVID) when analysing total number of tackles, ball carries, and tackler events per match (i.e., summated total number of tackle events during a match) (Table 3). However, there was a significant increase in the total number of tackles and tackler involvements when comparing season 2020_b (post-COVID) to all other seasons. This finding may be explained because of rule changes implemented at the commencement of season 2020_b. Following a 141-day suspension of fixtures due to COVID-19 restrictions,[12] competitive SL matches returned with the removal of the scrum and the introduction of the "six-again" rule. This rule change allowed the referee to award six more tackles in the event of an infringement by a defending side. As such governing bodies should be aware of the impact rules changes have on match activities, specifically tackle events, given their injury[11] and concussion risk.[27] Gardner[28] demonstrated that, during NRL match play, 94.5 % of head injury assessments (HIAs) occurred during the tackle event and reported 1.6 HIAs per 1000 tackler involvements. Considering the number of tackler involvements reported within the present study, it would be reasonable to speculate that the risk of HIA during rugby league match-play has increased from a propensity of 1.5 HIAs per match in season 2014 (964 tackler involvements per match) to a propensity of 1.7 HIAs per match in season 2020_b (1068 tackler involvements per match); however further research is required to substantiate this speculation.

The majority of studies within rugby league utilise small sample sizes (i.e., single team[7]) during data collection which is considered as a limitation.[29] In contrast, the large number of players and matches is a strength of the present study (e.g., previous large sample studies have included data from 380 players collected from 323 matches during a two-year period vs. the present study included >1,000,000 tackle events and data from 864 players collected from 1176 matches during a seven-year period). This study is not without its limitations. Firstly, the specific action of the tackler (e.g., type of tackle completed) during the tackle event is not specified therefore the tackling player may be the tackler making initial contact or perform the role of a supporting tackler. Secondly, the tackle events reported do not provide a

Table 3
Total number (mean ± 95 % CIs) of tackle events, ball carries and tackler involvements per match.

Season	Tackle events	Ball carries (i.e., attacker)	Tackler involvements (i.e., defender)
2014	967 (956–977)[b, c, d, e, f, g, h]	303 (299–306)[h]	664 (657–671)[b, c, d, e, f, g, h]
2015	994 (969–1020)[h]	304 (295–312)[h]	691 (672–709)[a, h]
2016	999 (973–1025)[h]	299 (291–308)[h]	700 (681–718)[a, h]
2017	1002 (977–1028)[h]	305 (296–313)[h]	698 (680–716)[a, h]
2018	999 (974–1024)[h]	299 (291–307)[h]	700 (682–718)[a, h]
2019	997 (971–1022)[h]	300 (292–308)[h]	697 (679–715)[a, h]
2020_a	1012 (975–1048)[h]	305 (293–317)	707 (681–732)[a, h]
2020_b	1068 (1037–1099)[a, b, c, d, e, f, g]	318 (308–328)[a, b, c, d, e, f]	750 (728–772)[a, b, c, d, e, f, g]

Values are presented as mean (95 % CI); CI = confidence interval.
[a] Significantly different from 2014.
[b] Significantly different from 2015.
[c] Significantly different from 2016.
[d] Significantly different from 2017.
[e] Significantly different from 2018.
[f] Significantly different from 2019.
[g] Significantly different from 2020a.
[h] Significantly different from 2020b.

magnitude of the collision. Collisions of different intensity are likely to have different metabolic (e.g., fatigue) or mechanical (e.g., muscle damage) costs. One way to examine collision intensity, specifically at the head, is in the form of head acceleration magnitudes and exposures, which has the potential to be achieved using instrumented mouthguards,[30] however this method of collision quantification remains in its infancy.

5. Conclusion

For the first time, this study quantifies the highest number of tackle events, ball carry, and tackler involvements within rugby league match-play at various time epochs ranging from 1-min to whole-match. Whilst whole-match analysis shows that players are exposed to one tackle event every 93 to 259 s, peak 1-min analysis reveals an exposure at a substantially greater rate of every 17 to 29 s. This is an important consideration since during periods of a greater tackler involvement rate, players will experience fatigue which may compromise tackle technique and therefore it is important that players prepare physically to develop the capacity to perform to these demands. This study demonstrated low variability (CV <10 %) within-player, between-matches and between-seasons for tackles, ball carries, and tackler involvements at all time epochs which suggests a consistent exposure to these events. This enables practitioners to understand when a player has been exposed to a meaningful change in collision stimulus and allows for simple tackle exposure modification via match selection/deselection or rest. Matches completed following a break in fixtures due to COVID-19 restrictions reported significantly greater total number tackle and tackler involvements when analysed at a match level.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2022.07.002.

Funding Information

No external sources of funding were obtained for this study.

Declaration of Interest Statement

The authors report no conflict of interests.

Confirmation of Ethical Compliance

Institutional ethics approval was granted by Leeds Beckett University Human Ethics Research Committee. Application Ref: 98724.

Acknowledgements

No external sources of funding were obtained for this study. The authors have no conflict of interests to declare. The authors are grateful to Opta for their support. The authors would also like to express their gratitude to coaches, practitioners and players within all Super League clubs who supported the project.

References

1. Hausler J, Halaki M, Orr R. Application of global positioning system and microsensor technology in competitive rugby league match-play: a systematic review and meta-analysis. *Sports Med* 2016;46(4):559-588. doi:10.1007/s40279-015-0440-6.
2. Johnston RD, Weaving D, Hulin BT et al. Peak movement and collision demands of professional rugby league competition. *J Sports Sci* 2019;37(18):2144-2151. doi:10.1080/02640414.2019.1622882.
3. Weaving D, Sawczuk T, Williams S et al. The peak duration-specific locomotor demands and concurrent collision frequencies of European Super League rugby. *J Sports Sci* 2019;37(3):322-330. doi:10.1080/02640414.2018.1500425.
4. Tee JC, Till K, Jones B. Incidence and characteristics of injury in under-19 academy level rugby league match play: a single season prospective cohort study. *J Sports Sci* 2019;37(10):1181-1188. doi:10.1080/02640414.2018.1547100.
5. Naughton M, Jones B, Hendricks S et al. Quantifying the collision dose in Rugby league: a systematic review, meta-analysis, and critical analysis. *Sport Med - Open* 2020;6(1):6. doi:10.1186/s40798-019-0233-9.
6. Dalton-Barron N, Whitehead S, Roe G et al. Time to embrace the complexity when analysing GPS data? A systematic review of contextual factors on match running in rugby league. *J Sports Sci* 2020;38(10):1161-1180. doi:10.1080/02640414.2020.1745446.
7. Glassbrook DJ, Doyle TLA, Alderson JA et al. The demands of professional Rugby league match-play: a meta-analysis. *Sport Med - Open* 2019;5(1):24. doi:10.1186/s40798-019-0197-9.
8. Rennie G, Dalton-Barron N, McLaren SJ et al. Locomotor and collision characteristics by phases of play during the 2017 rugby league World Cup. *Sci Med Footb* 2019;00(00):1-8. doi:10.1080/24733938.2019.1694167.
9. Fitzpatrick AC, Naylor AS, Myler P et al. A three-year epidemiological prospective cohort study of rugby league match injuries from the European Super League. *J Sci Med Sport* 2018;21(2):160-165. doi:10.1016/j.jsams.2017.08.012.
10. Hulin BT, Gabbett TJ, Johnston RD et al. Wearable microtechnology can accurately identify collision events during professional rugby league match-play. *J Sci Med Sport* 2017;20(7):638-642. doi:10.1016/j.jsams.2016.11.006.
11. Hopkinson M, Nicholson G, Weaving D et al. Rugby league ball carrier injuries: the relative importance of tackle characteristics during the European Super League. *Eur J Sport Sci* 2020;0(0):1-20. doi:10.1080/17461391.2020.1853817.
12. Rennie G, Hart B, Dalton-Barron N et al. Longitudinal changes in Super League match locomotor and event characteristics: a league-wide investigation over three seasons in rugby league. *PLoS One* 2021;16(12):e0260711. doi:10.1371/journal.pone.0260711.
13. Delves RIM, Thornton HR, Hodges J et al. The introduction of the six-again rule has increased acceleration intensity across all positions in the National Rugby League competition. *Sci Med Footb* 2022;00(00):1-10. doi:10.1080/24733938.2022.2051729.
14. Jones B, Phillips G, Kemp S et al. SARS-CoV-2 transmission during rugby league matches: do players become infected after participating with SARS-CoV-2 positive players? *Br J Sports Med* 2021. doi:10.1136/bjsports-2020-103714. (October 2020): bjsports-2020-103714.
15. Emmonds S, O'Hara J, Till K et al. Physiological and movement demands of rugby league referees. *J Strength Cond Res* 2015;29(12):3367-3374. doi:10.1519/JSC.0000000000001002.
16. Evans SD, Brewer C, Haigh JD et al. The physical demands of super league rugby: experiences of a newly promoted franchise. *Eur J Sport Sci* 2015;15(6):505-513. doi:10.1080/17461391.2015.1041064.
17. Parmar N, James N, Hughes M et al. Team performance indicators that predict match outcome and points difference in professional rugby league. *Int J Perform Anal Sport* 2017;17(6):1044-1056. doi:10.1080/24748668.2017.1419409.
18. Woods CT, Robertson S, Sinclair WH et al. Non-metric multidimensional performance indicator scaling reveals seasonal and team dissimilarity within the National Rugby League. *J Sci Med Sport* 2018;21(4):410-415. doi:10.1016/j.jsams.2017.06.014.
19. Opta. *Rugby league post match analysis operational definitions, Second.*, Opta, 2013.
20. Ushey K. Package 'RcppRoll'. Published online 2018:1–5: https://cran.r-project.org/web/packages/RcppRoll/RcppRoll.pdf.
21. Bates D, Mächler M, Bolker B et al. Fitting linear mixed-effects models using lme4. *J Stat Softw* 2015;67(1). doi:10.18637/jss.v067.i01.
22. Howe ST, Aughey RJ, Hopkins WG et al. Sensitivity, reliability and construct validity of GPS and accelerometers for quantifying peak periods of rugby competition. *PLoS One* 2020;15(7):1-23. doi:10.1371/journal.pone.0236024.
23. Dalton-Barron N, Palczewska A, McLaren SJ et al. A league-wide investigation into variability of rugby league match running from 322 Super League games. *Sci Med Footb* 2021;05(03):225-233. doi:10.1080/24733938.2020.1844907.
24. Tierney P, Blake C, Delahunt E. The relationship between collision metrics from micro-sensor technology and video-coded events in rugby union. *Scand J Med Sci Sports* 2020;30(11):2193-2204. doi:10.1111/sms.13779.
25. Austin D, Gabbett T, Jenkins D. Tackling in a professional rugby league. *J Strength Cond Res* 2011;25(6):1659-1663. doi:10.1519/JSC.0b013e3181da781c.
26. Hogarth LW, Burkett BJ, McKean MR. Match demands of professional rugby football codes: a review from 2008 to 2015. *Int J Sports Sci Coach* 2016;11(3):451-463. doi:10.1177/1747954116645209.
27. Gardner A, Iverson GL, Levi CR et al. A systematic review of concussion in rugby league. *Br J Sports Med* 2015;49(8):495-498. doi:10.1136/bjsports-2013-093102.
28. Gardner AJ, Iverson GL, Edwards S et al. A case–control study of tackle-based Head Injury Assessment (HIA) risk factors in the National Rugby League. *Sport Med - Open* 2021;7(1):84. doi:10.1186/s40798-021-00377-9.
29. Weaving D, Jones B, Ireton M et al. Overcoming the problem of multicollinearity in sports performance data: a novel application of partial least squares correlation analysis. *PLoS One* 2019;14(2):e0211776. doi:10.1371/journal.pone.0211776.
30. Tierney G, Weaving D, Tooby J et al. Quantifying head acceleration exposure via instrumented mouthguards (iMG): a validity and feasibility study protocol to inform iMG suitability for the TaCKLE project. *BMJ Open Sport Exerc Med* 2021;7(3):e001125. doi:10.1136/bmjsem-2021-001125.